

How do humans learn?

Biological Foundations

SYS 478. Fall 2026. Prof. Sarah Van Wart



Agenda

- Intelligence & learning discussion
- Nervous system + brain
- Neurons + networks
- Plasticity

Activity 1: Discuss at your table...

1. What is intelligence?
 - a. Are trees intelligent?
 - b. Is DNA intelligent?
2. What is learning?
 - a. Do trees learn?
3. How would you know that learning has happened?
4. Can something be intelligent that can't learn?
5. Can something learn without being intelligent?

What is intelligence?

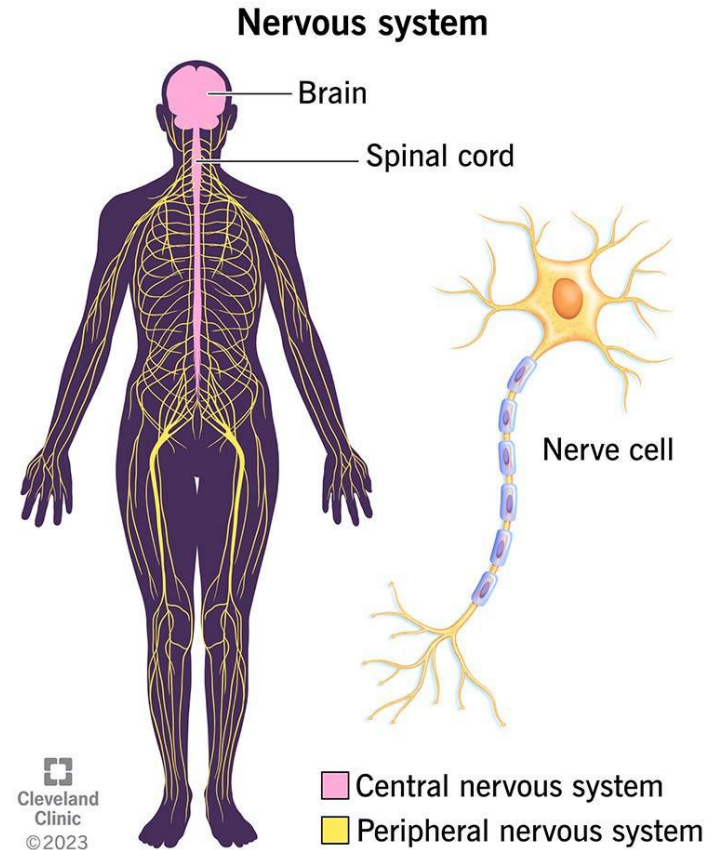
1. Ability to reason (?) and solve problems
2. Not quite as sophisticated as having 'consciousness'
3. Trees:
 - a. Adapt
4. Types: emotional, physical, spiritual, spatial
5. Machines:
 - a.

What is learning?

1. Process of gaining knowledge
2. ...and apply it to new situations
3. Gaining and retaining
- 4.

The Nervous System

- It coordinates perception, movement, memory, emotion, and behavior.
- Made up of the brain, spinal cord, and peripheral nervous system.
- It receives information from inside and outside the body.
- Sends signals between body and brain, passed along nerve cells

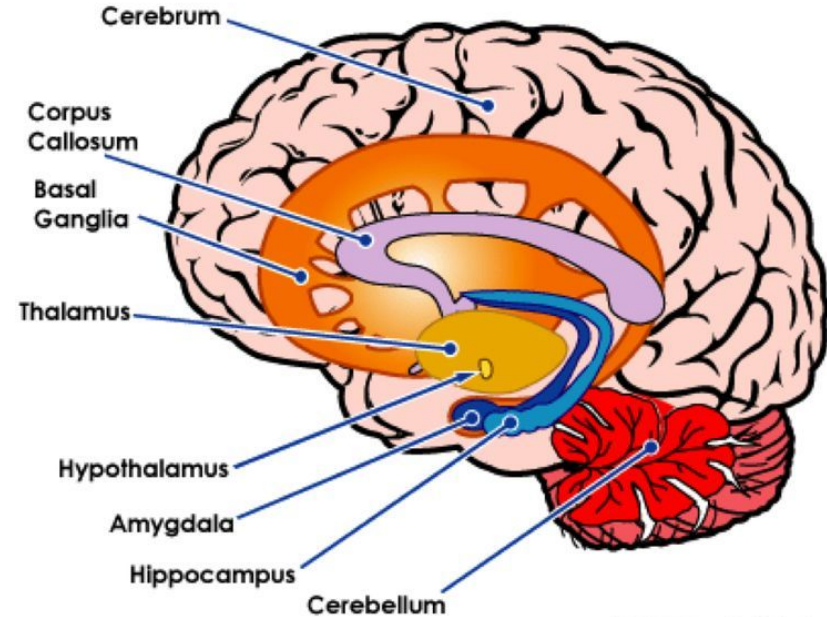


The Brain Is a Specialized, Interconnected System

This is an oversimplification, but...

- **Cortex:** perception, language, planning, reasoning
- **Hippocampus:** formation of new memories
- **Amygdala:** emotional significance and associative learning
- **Basal ganglia** – action, habits, reward-based learning
- **Cerebellum** – coordination, prediction, motor learning
- **Brainstem** – basic bodily functions

Basal Ganglia and Limbic System



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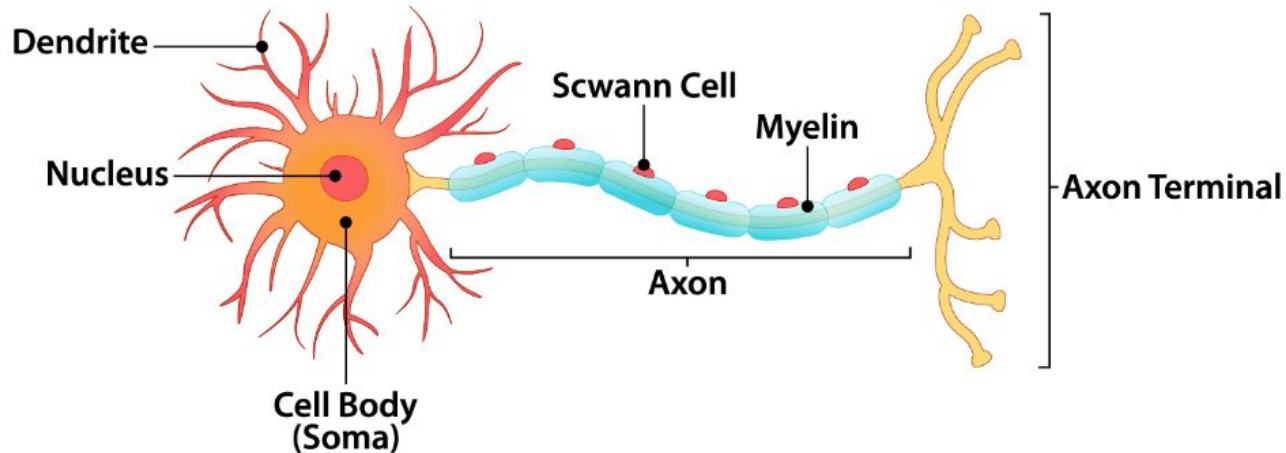
Brain Facts

- ~86 billion neurons, with roughly 1 quadrillion connections known as synapses wiring these cells together. (That is 1 million billion synapses)
- 80% of neurons are in the cerebellum (just 10% of the brain's mass)
- Lots of “storage” space – “2.5 petabytes” (but difficult to make direct comparisons to traditional storage media.
- Neurons are very dense – A piece of brain tissue the size of a grain of sand contains 100,000 neurons and 1 billion synapses.
- The human brain is only about 2% of body weight but uses roughly 20% of the body's metabolic energy.
- Even so, the brain operates on only about 20 watts (1 hour of power to a small LED light bulb).

Zooming In: The Neuron

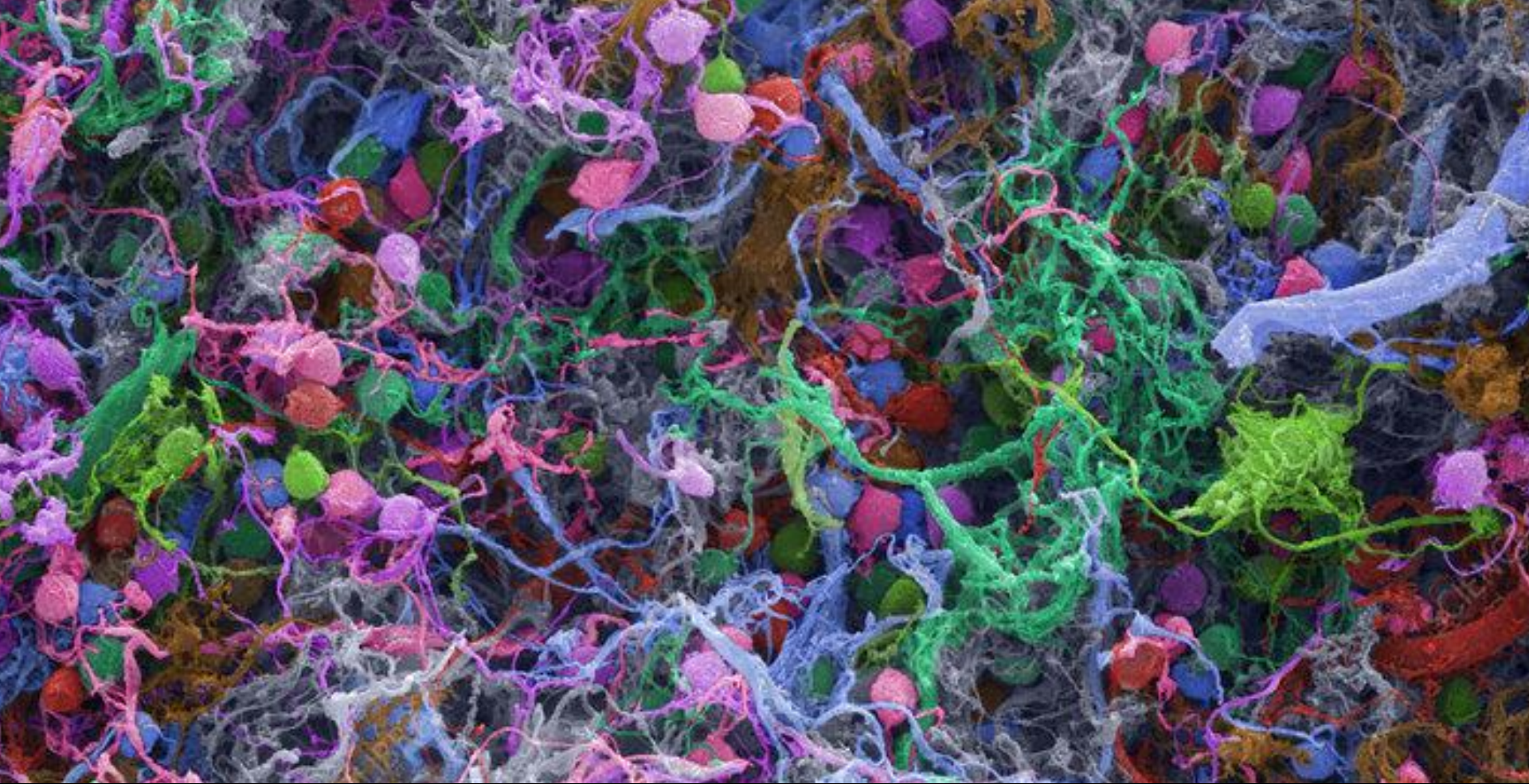
Neurons are specialized cells that, receive signals, integrate information, generate electrical signals, and communicate with other cells

A living cell, not simply a wire.



Neurons Work in Networks

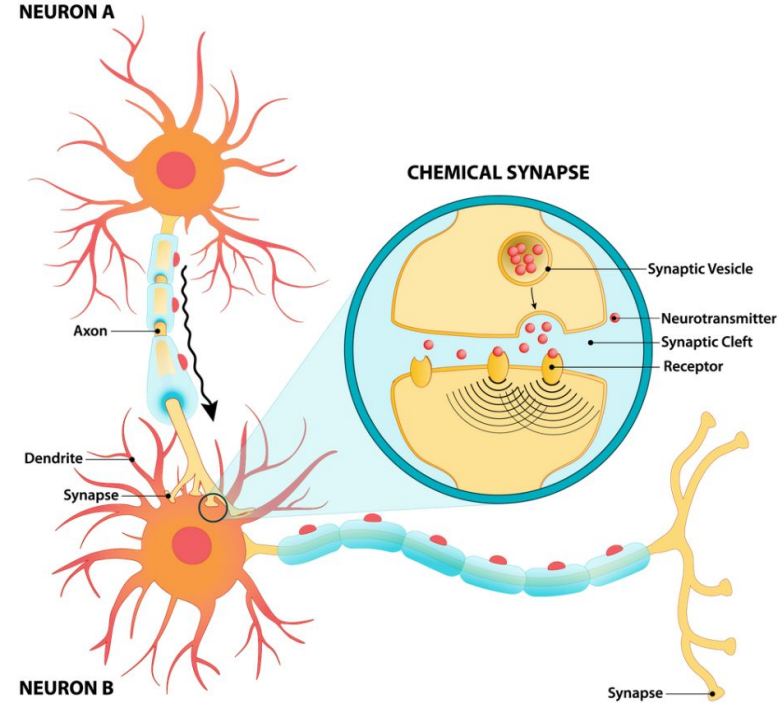
- A single neuron may receive input from thousands of other neurons.
- Neurons influence one another through networks of connections.
- Patterns of activity across many neurons contribute to perception and behavior.
- What matters is not only individual neurons, but their connections and patterns of activity.



Source: <https://www.sciencephoto.com/media/776542/view/human-brain-cells-sem> (Magnification x700)

Neuron Communication

- **Synapses:** Connections between neurons; Where axon terminals meet the dendrites or cell body of a neuron.
- **Neurotransmitters:** chemicals that get emitted that can stimulate an action potential (as sodium ions (Na^+) rush into the neuron)
- **Action Potential:** After a charge threshold is reached, a rapid electrical impulse that travels down the neuron and sends neurotransmitters to connected neurons.



Electrical Signaling Within a Neuron

Neurons maintain an electrical difference across their membranes.

When incoming activity is sufficient:

- the neuron's membrane reaches a threshold
- an action potential occurs
- the electrical signal travels down the axon

[Demo](#)

Connections Are Not All Equally Strong

One neuron can affect another one:

- strongly
- weakly
- excitatorily
- inhibitory

The effect of a connection can also change over time.

This is known as “**Synaptic Plasticity**” – the capacity of connections between neurons to change.

Learning Physically Changes the Nervous System

Experience can produce changes in:

- synaptic strength
- numbers or structures of synapses
- patterns of neural activity
- responsiveness of neurons and networks

These changes can alter how the system responds to future experiences.

Experience → neural activity → biological change → different future response

That could be your major conceptual diagram for the lesson.

“Cells That Fire Together Wire Together”

When the same neurons repeatedly engage through experience, the connections between them can sometimes become stronger.

This is known as **Hebbian Learning**: “Cells that fire together wire together.”

But:

- this is an intuition, not a complete rule
- connections can also weaken
- timing matters
- context matters
- different neural systems use different forms of plasticity

THE **BRIGHT** AND **DARK** SIDES OF NEUROPLASTICITY



Neuroplasticity makes
your brain resilient.

Neuroplasticity enables you to
recover from stroke, injury,
and birth abnormalities.

You can learn new ways
of being and responding
to conflict.

In many cases, you can also
overcome depression, addiction,
obsessive compulsive patterns,
ADHD, and other issues.



Neuroplasticity means
the brain is always learning.

But the brain is neutral -
it doesn't know the difference
between good and bad.

It learns whatever is repeated -
both helpful and unhelpful
thoughts, actions, and habits.

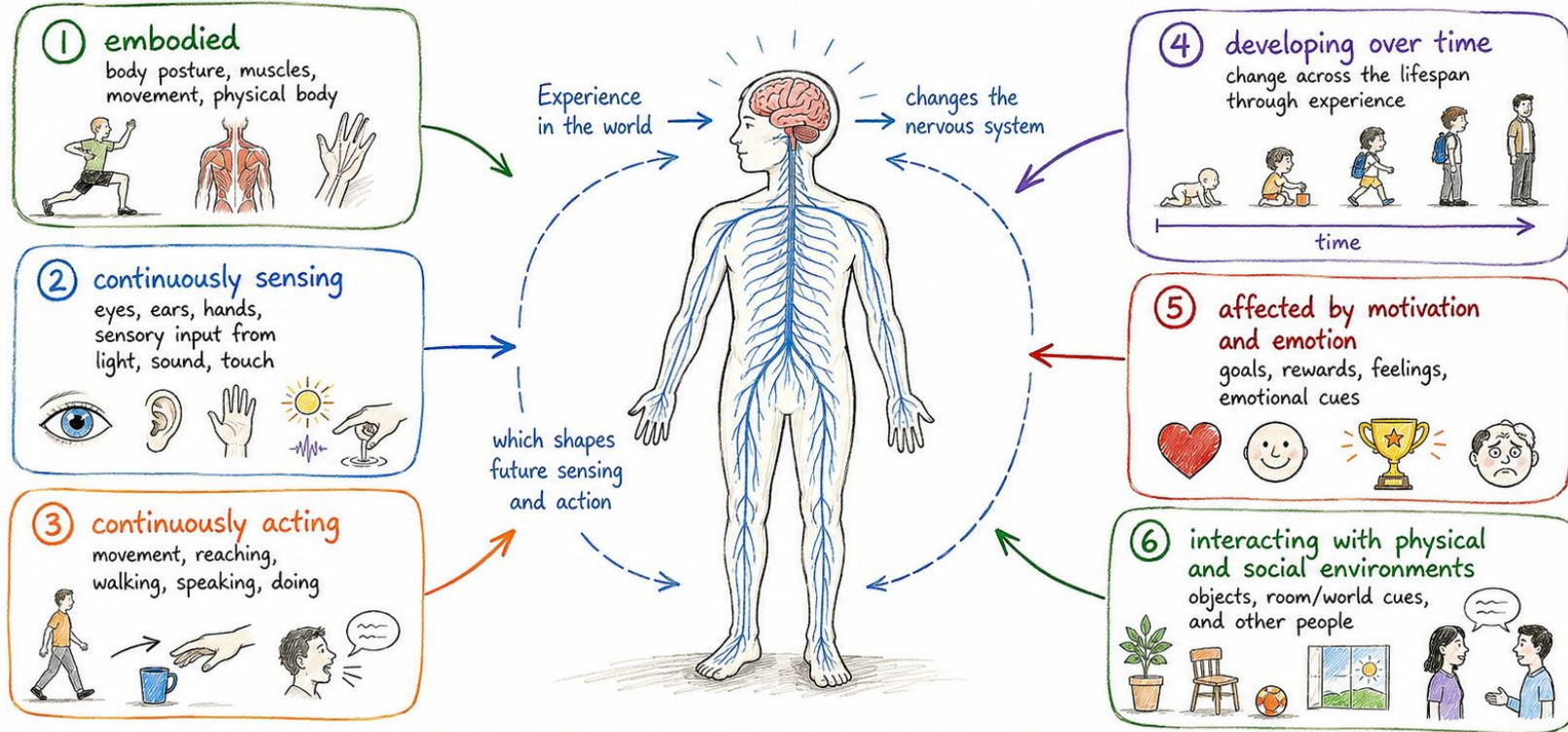
Therefore neuroplasticity may
entrench depressive, anxious,
obsessive, and over-reactive
patterns.

Learning

Activity: Forms of Learning

- [Discuss at your tables...](#)
- Your thoughts here...

A Nervous System Learns While Living in a World



Biological learning is highly dependent on experience and environmental conditions.

One more set of questions: what is a “self”?

- If the self depends entirely on neurons and synapses, what exactly is the “self”? In other words, if our memories, emotions, and personality can change when the brain changes, what makes us the same person over time?
- What would count as evidence that another system – biological or artificial – is actually conscious rather than merely behaving as if it were?
- If a model were able to replicate all of the neurons and synaptic connections / weights, would that be the same as a person?

In what ways do Als mimic biology?

A Question We'll Return To

Computer scientists later borrowed ideas like **networks**, **connections**, and **changing connection strengths** (synaptic plasticity) when developing artificial neural networks.

But biological nervous systems and artificial neural networks are ***not the same thing***

What is a Neural Network?